

National Academy of Science and Technology

(Affiliated to Pokhara University)

Dhangadhi, Kailali

Pre University Examinations

Level: Bachelor

Semester: VI_Spring

Year : 2025

Program: B.E. Computer

F.M. : 100

Course: Image Processing and Pattern Recognition

P.M. : 45

Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Digital Image Processing? Explain the Basic Phases of Digital Image Processing. 7
b) What are applications of histogram? Equalize the following histogram 8
Gray scale= [0,7]

2	3	3	2
4	2	4	3
3	2	3	5
2	4	2	4

2. a) Explain the image enhancement technique using magnification with replication and interpolation? 7
b) Explain DFT Transform and find the phase angle, power spectrum and Fourier spectrum at point 0,1 of given image? 8

$$F(x) \begin{bmatrix} 0 \\ 2 \\ 4 \\ 6 \end{bmatrix}$$

"OR"

3. a) Explain Haar Transform with algorithm and its properties. 8
b) Explain Image degradation and Restoration Model with its Importances. 8
b) Differentiate between Lossless and Lossy Compression Techniques with neat diagram too. 7
4. a) Explain Image segmentation techniques based on Region Growing, splitting and merging. 8
b) Define Image Segmentation. Explain its types based on Dissimilarities Properties. 7

5. a) Construct Huffman tree for each gray level given and find the efficiency. 8

Gray Level	0	1	2	3	4	5	6	7
No. of pixels	450	1250	650	2034	815	1560	350	1400

- b) Find the Shape Number for following Image. 7

			A			
		A		A		
	A				A*	
A				A		
A					A	A
A						A
	A				A	
		A	A		A	
			A			A
				A	A	

6. a) Explain Artificial Neural Network. Explain Hopfield Network in detail. 7
b) Define Pattern and Pattern Class. Explain Steps in Pattern Recognition. 8

"OR"

Explain SIFT and PCA in detail.

7. Write short notes on following (Any Two) 5x2
a) Sampling and Quantization
b) Run Length Encoding
c) Fast Fourier Transform

Good Luck!!!